

Sutherland-Hodgeman Polygon Clipping:

Polygon :

A Polygon can be described as the enclosed collection or group of the lines.”

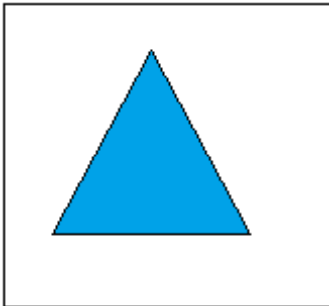
In a polygon, all lines are connected. Lines can be a combination of edges and vertices, which together form a polygon. A polygon refers to a two-dimensional architecture made up of a number of straight lines.

Some Examples of the polygon:

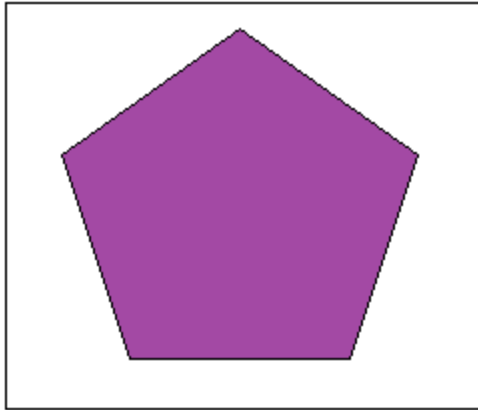
1. Triangles
2. Pentagons
3. Hexagons
4. Quadrilaterals

The polygon's name defines how many sides the architecture contains.

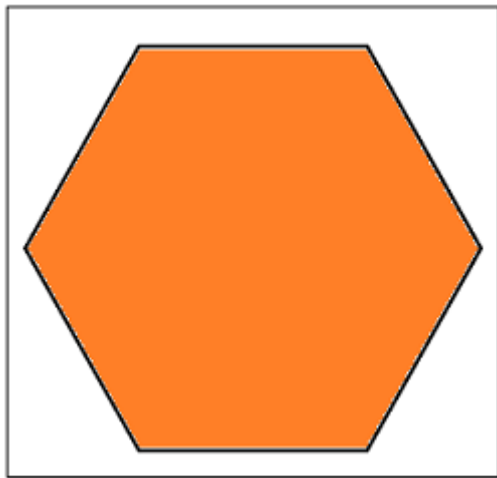
Triangle: It has three sides.



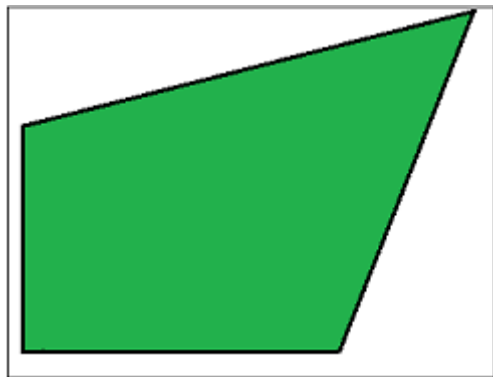
Pentagon: A pentagon has five sides.



Hexagon: It contains six sides.



Quadrilaterals: It contains four sides.



What is clipping?

Generally, any procedure that identifies those portions of a picture that are either inside or outside of a specified region of a space is referred to as a clipping algorithm or simply clipping.

Polygon Clipping

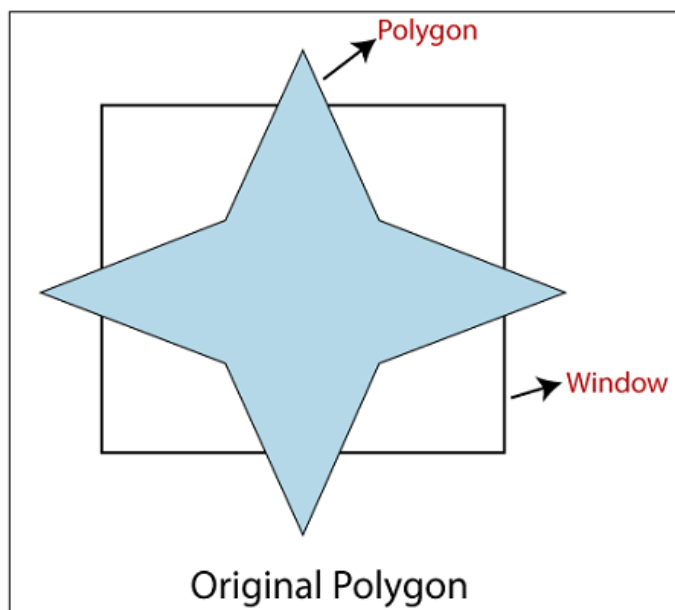
Polygon clipping is a process in which we only consider the part which is inside the view pane or window. We will remove or clip the part that is outside the window. We will use the following algorithms for [polygon clipping](#)–

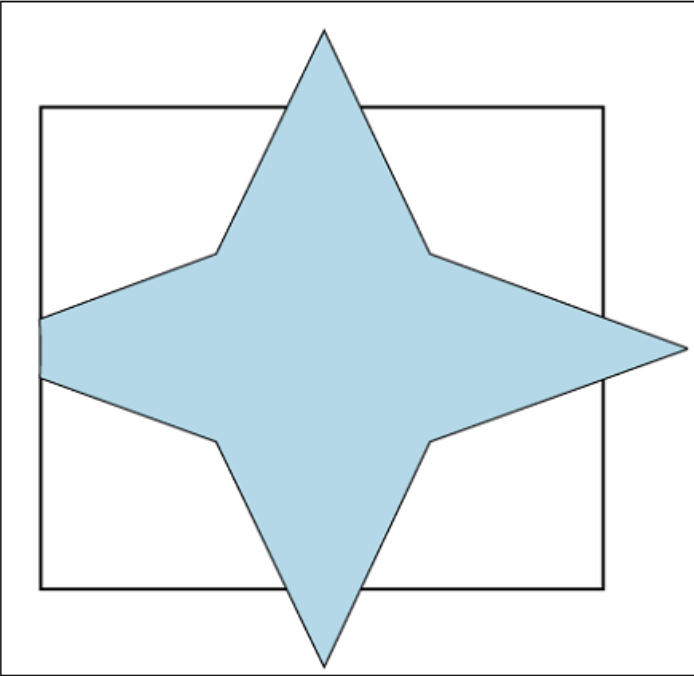
- Sutherland-Hodgeman polygon clipping algorithm
- Weiler-Atherton polygon clipping algorithm

Sutherland-Hodgeman polygon clipping algorithm

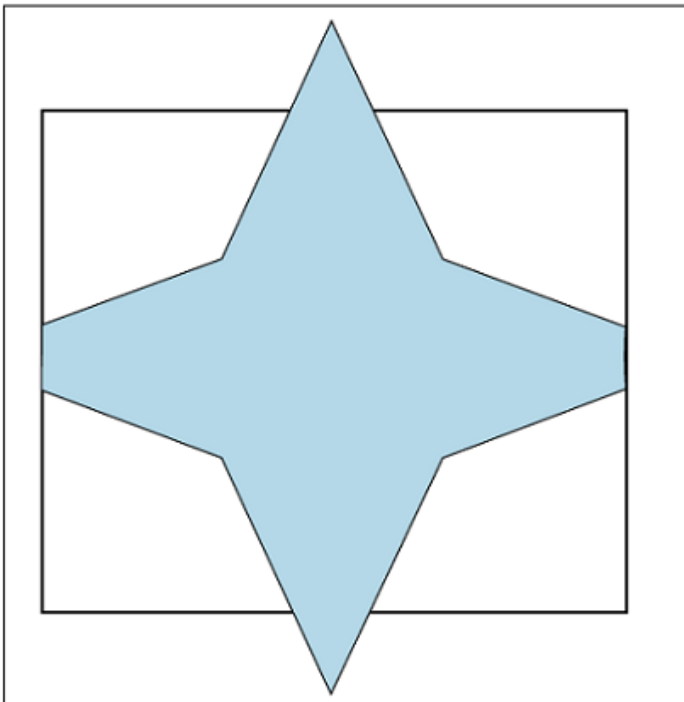
The polygon clipping algorithm deals with four different clipping cases. The output of each case is input for the next case.

Case1) Left clip: In the left side polygon clipping, we only remove the left part of the polygon, which is outside the window. We only save the portion which is inside the window.

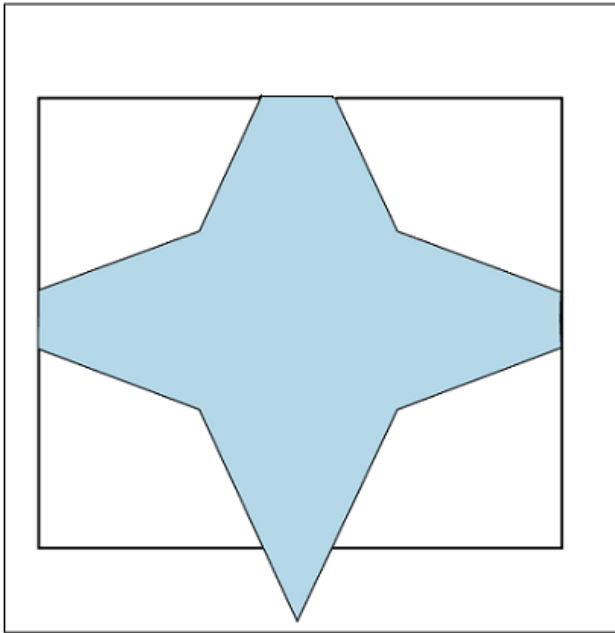




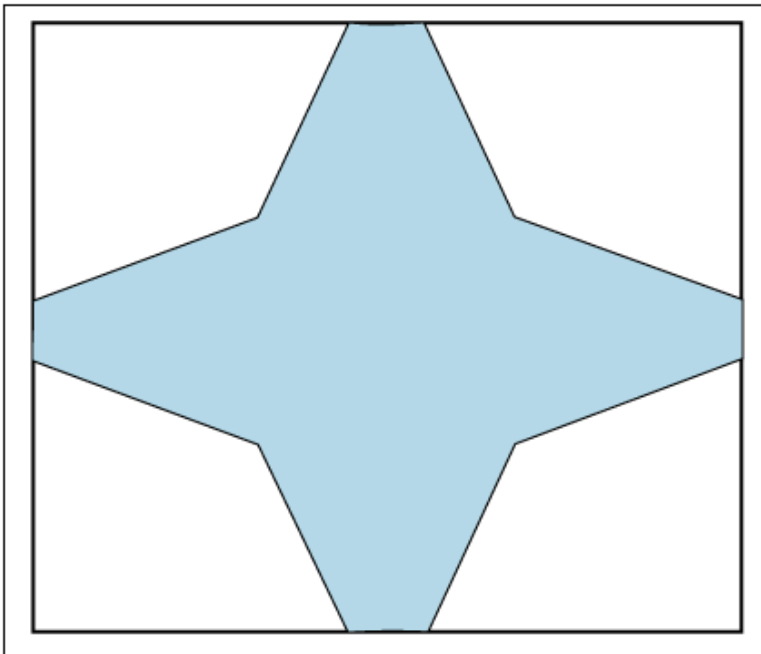
Case2) Right clip: In the right-side polygon clipping, we only remove the right part of the polygon, which is outside the window. We only save the portion which is inside the window.



Case3) Top clip: On the top side polygon clipping, we only remove the top part of the polygon, which is outside the window. We only save the portion which is inside the window.

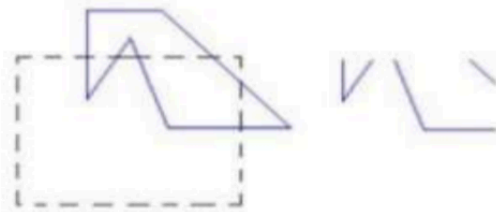


Case4) Bottom clip: In the bottom side polygon clipping, we only remove the bottom part of the polygon, which is outside the window. We only save the portion which is inside the window.



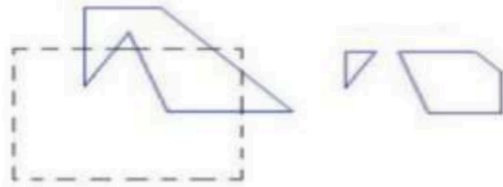
Let us understand this clipping method - • Example-

*To clip a **polygon**, we cannot directly apply a line-clipping method to the individual polygon edges because this approach would produce a series of **unconnected line segments** as shown in figure .*



The clipped polygons must be a bounded area after clipping as shown in figure.

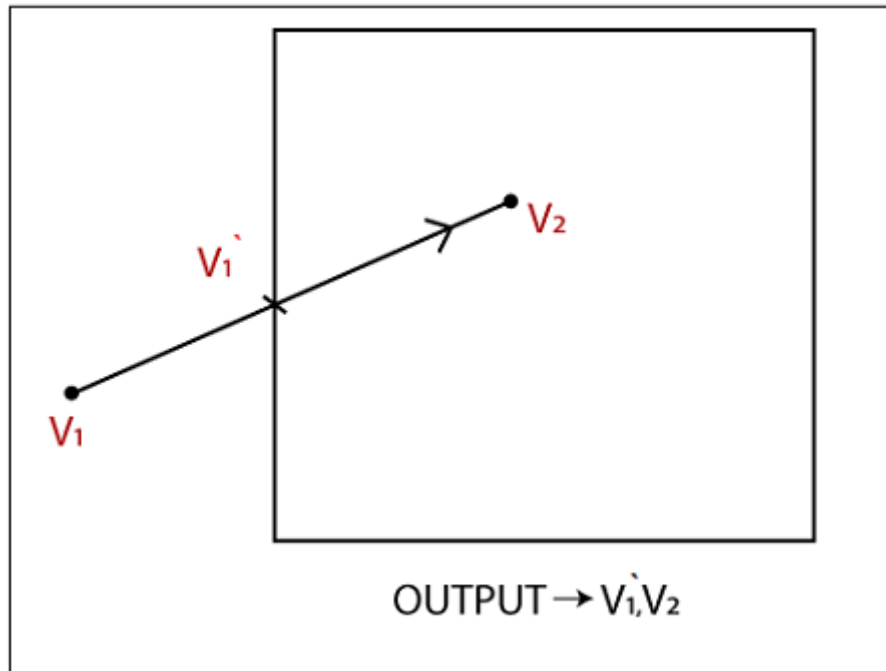
Concave Polygon



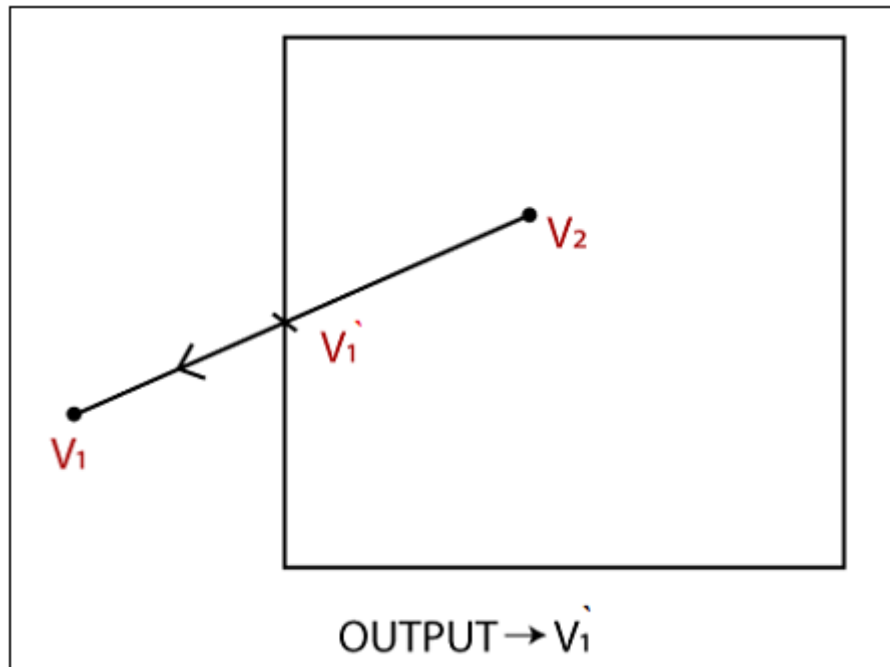
- For polygon clipping, we require an algorithm that will generate one or more **closed areas** that are then scan converted for the area fill.
- The output of a polygon clipper should be a **sequence of vertices** that defines the clipped polygon boundaries.

There should be the following conditions while we clip a polygon.

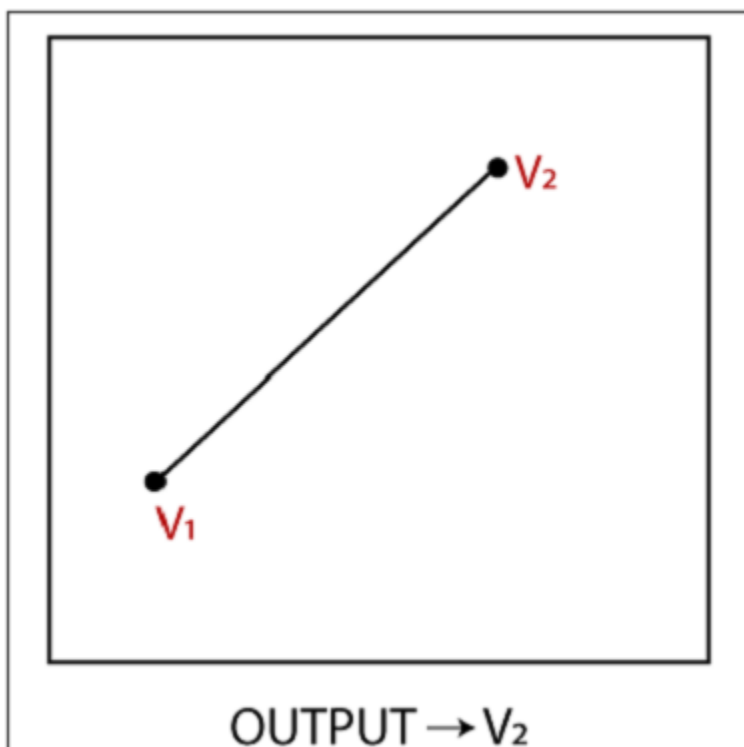
Condition 1(Out-In): If the first vertex of the polygon is outside and the second vertex is inside the window, then the output will be the intersection point and second vertex.



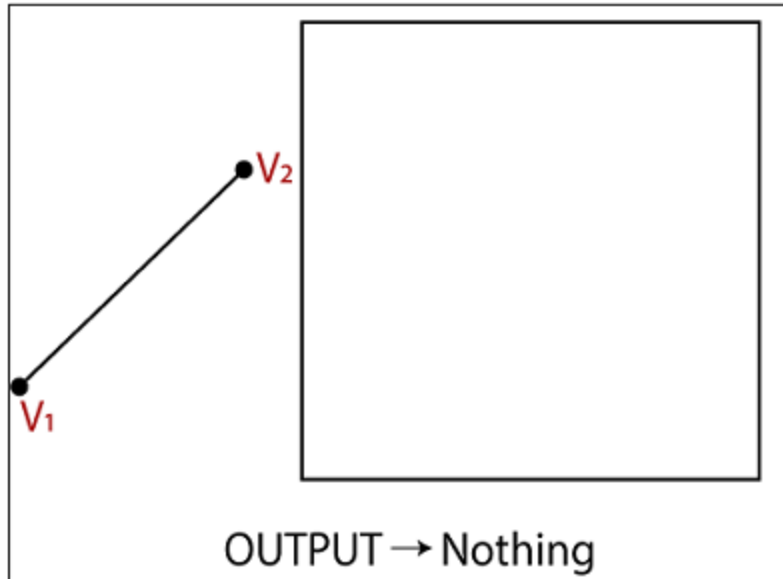
Condition 2(In-Out): If the first vertex of the polygon is inside and the second vertex is outside the window, then the output will be the intersection point.



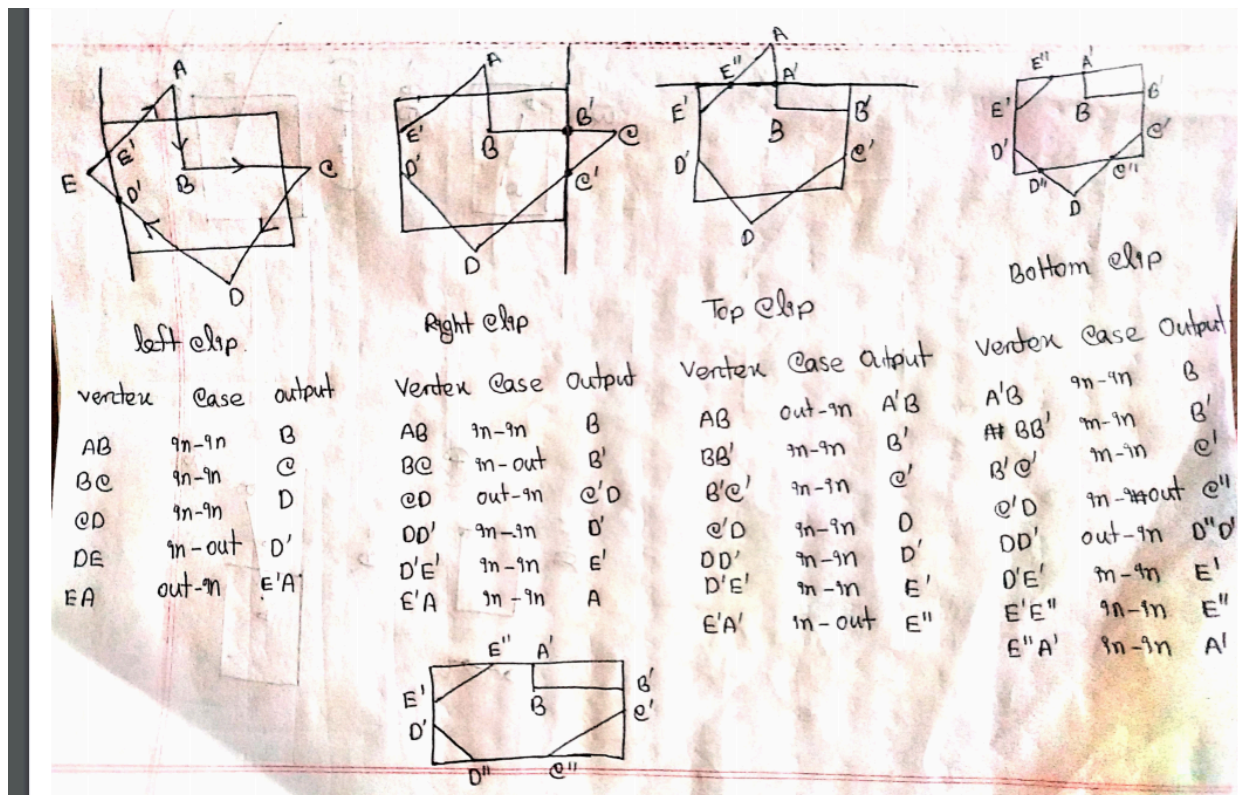
Condition 3(In-In): If both vertices of the polygon are inside the window, then the output will be the second vertex.



Condition 4(Out-Out): If both vertices of the polygon are outside the window, then the output will be nothing. It means we found a clipped polygon.



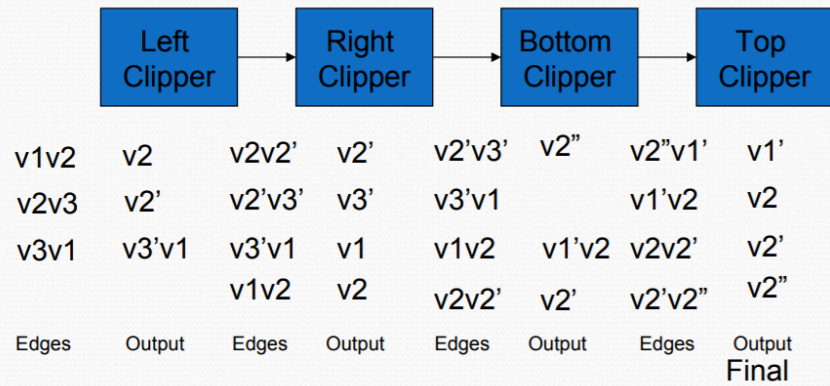
Example 1 :



Example 2 :

A diagram showing a triangle with vertices v_1 , v_2 , and v_3 . The dual triangle is formed by vertices v_1' , v_2' , and v_3' , which are the intersections of the lines containing the edges of the original triangle. Specifically, v_1' is the intersection of the lines containing edges v_2v_3 and v_3v_1 , v_2' is the intersection of the lines containing edges v_3v_1 and v_1v_2 , and v_3' is the intersection of the lines containing edges v_1v_2 and v_2v_3 . The dual triangle is shaded in light blue.

Figure 6-27, page 332



Disadvantage of Cohen Hodgmen Algorithm:

This method requires a considerable amount of memory. The first of all polygons are stored in original form. Then clipping against left edge done and output is stored. Then clipping against right edge done, then top edge. Finally, the bottom edge is clipped. Results of all these operations are stored in memory. So wastage of memory for storing intermediate polygons.